

Objectives

1. the students will be able to appreciate the historical background/perspective, from Hippocrates, to A.T. Still, to consideration of the 5 (or 6) Osteopathic Models, to Dr. Phil Greenman's legacy, and on to the related to inter-professional and global awareness of the Upper (and Lower) Crossed syndrome, from the legacy of neurologists, Drs. Janda and Lewit.
2. the students will be able to understand the Upper Crossed Syndrome, including the role of the type I (slow twitch, postural) and type II (fast twitch, phasic) muscles. The student will understand which muscles need stretching, and which muscles need strengthening, in relation to application of an understanding of this "crossed" syndrome. Specific exercises will be reviewed at the end of the lecture, with pictures and diagrams included.
3. the students will be able to understand the importance of obtaining and considering the patient's history (subjective), along with objective information. This objective information can include Special Tests, muscle testing/palpation, the osteopathic structural examination, postural observation and analysis, and other neurological (NCV / EMG) or radiological (x-rays, MRI, etc.) testing, as needed.
4. the students will be able to understand how to incorporate an understanding of postural compensation, tensegrity (a.k.a. biotensegrity), gravitational strain pathophysiology (G.S.P.) and postural decompensation into their osteopathic perspective.
5. the students will read the assigned article from Page (LINK: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3201065/>), which is also listed under the Resources for this Classroom Session) on their own, to gain a sense of the clinical correlation / clinical suggestion related to this topic area.

Outline

- Historical perspective
- Review Muscle Types (I and II)
- Upper Crossed Syndrome
- Postural Considerations
 - Postural Compensation
 - Tensegrity
 - Gravitational Strain Pathophysiology (G.S.P.) and Postural Decompensation
- General Treatment Strategies

Historical Perspective

Vladimir Janda, M.D., D.Sc. (1928-2002)

- H/o poliomyelitis as a teenager, wheelchair-bound for 3 years after an initial quadriplegia.
- **Neurologist** training at Charles University, Prague.
- 1960s in Canada, he did post-doctoral work with Professor John Basmajian.
- His work in rehabilitation included research and clinical activities, over many decades and with a global influence.
- He was the Editor in Chief of the Czech Journal, Rehabilitation and Physical Medicine, until 2002.
- He worked with **Karel Lewit, M.D.**, another Czech **neurologist**, **presenting the Upper & Lower Cross Syndromes.**



Vladimir Janda, MD, DSc 1928–2002.

In clinical practice, we distinguish two muscle types in the evaluation and treatment:

**Type I muscles = postural muscles =
slow twitch muscles**

These muscles react by becoming TIGHT or SHORTENED as they become “facilitated” in response to postural imbalance, injury, somatic dysfunction.

Thus it is good to know which muscles are being categorized as postural/slow twitch muscles.

**Type II muscles = phasic muscles =
fast twitch muscles**

These muscles react by becoming WEAK as they become “inhibited,” For instance, the Gluteus medius muscle, a phasic muscle, may become reflexly inhibited due to an associated sacroiliac joint issue (e.g. extended, torsion, etc.) on one side and thus causing the muscle to become relatively weak on that side vs. the other side. The muscle itself has “not lost its innate strength” but rather is inhibited and thus becomes weak as a result.

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
TYPE I	Postural	Slow twitch	Become facilitated and thus SHORTENED or TIGHT	Upper crossed syndrome: Example 1: Head forward posture – <u>Anterior tight/shortened</u>= facilitated muscles: Pectorales <u>Posterior tight /shortened</u>=facilitated muscles: Levator scapulae, upper trapezius, SCM Example 2: when there is excessive ankle pronation – one can see the lower crossed syndrome – <u>Anterior: tight</u> iliopsoas muscle and <u>Posterior: tight</u> erector spinae muscles

Type I and Type II muscles in the clinical setting

Fiber Type	Postural or Phasic	Slow twitch or Fast twitch	Clinically – when involved – this muscle type tends to:	Examples in the clinical setting
Type II	Phasic	Fast twitch	Become <u>inhibited</u> and thus WEAK	Upper crossed syndrome (example: head forward posture) <u>Anterior: weak</u> neck flexors=inhibited <u>Posterior: weak</u> rhomboids/serratus anterior LOWER crossed syndrome: <u>Anterior: weak/inhibited</u> abdominals <u>Posterior: weak/inhibited</u> gluteus maximus/medius

Upper Crossed and Lower Crossed Syndromes

Clinically (e.g. work by Vladimir Janda, M.D./Prague/Czech Republic) there have been observed PATTERNS of muscle involvement, that have been called the **upper and lower crossed syndromes**. “Upper” because it involves the upper body, and “lower” because it involves the pelvis, or lower body.

It is called “Crossed” as there are two ‘lines’ that form a cross-over with ANTERIOR muscles being facilitated=tight=shortened (type I muscles) and POSTERIORLY (follow one LINE – call it the “facilitated” or “shortened” line) with type I muscles being also shortened. This is in response to a muscle imbalance.

Similarly, follow the other cross-line that goes from ANTERIOR to POSTERIOR, representing the INHIBITED and thus WEAK muscles both anteriorly and posteriorly.

So it is like a “puzzle” where you have two lines with four “stations:” 2 anterior and 2 posterior groups. Knowing the four “groups” – will give you an understanding of which muscles can be involved at any one time – so if the pectoralis major is tight in front, and you do have a cross syndrome – upper, for instance – you can deduce which other muscles are involved in a particular patient.

Table 1 – Anterior Muscles involved in Upper (and Lower) Crossed Syndromes

memorize

Anterior Postural Muscles	Anterior Phasic Muscles
Upper Cross	
Scalenes	Deep cervical flexors
Sternocleidomastoid	
Pectoralis major (upper)	
Pectoralis minor	
Lower Cross	
Iliopsoas	Rectus abdomini
Rectus femorus	Three Vasti muscles
Adductor brevis	Tibialis anterior

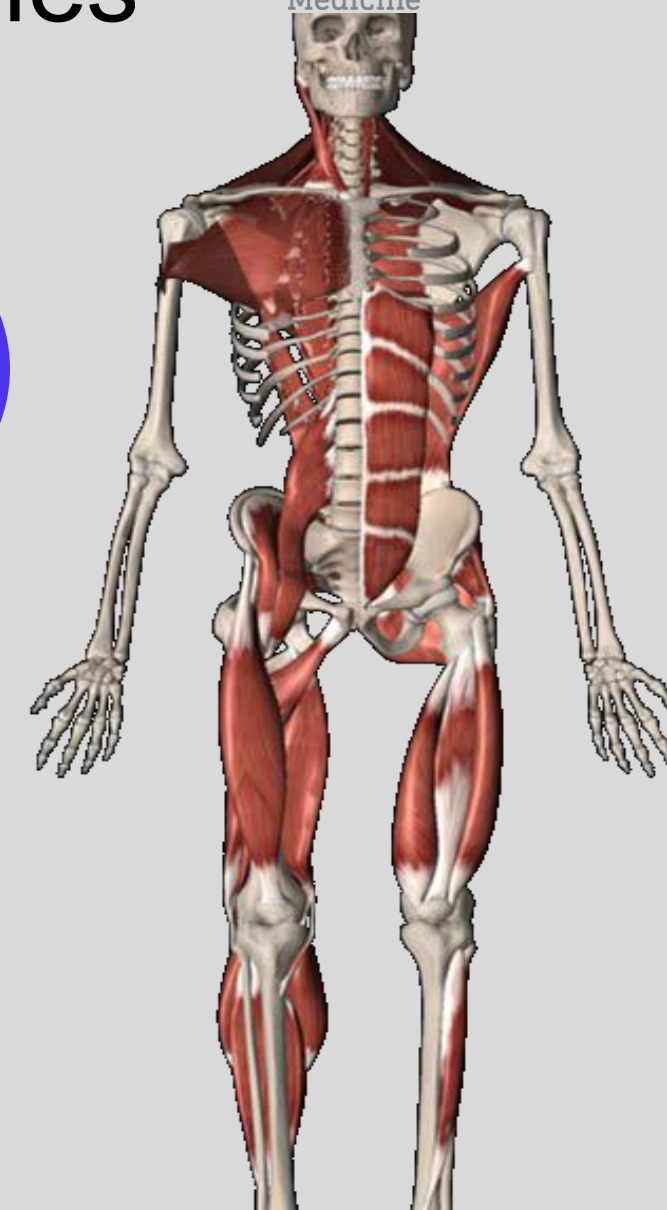


Table 2 – Posterior Muscles involved in Upper and Lower Crossed Syndromes

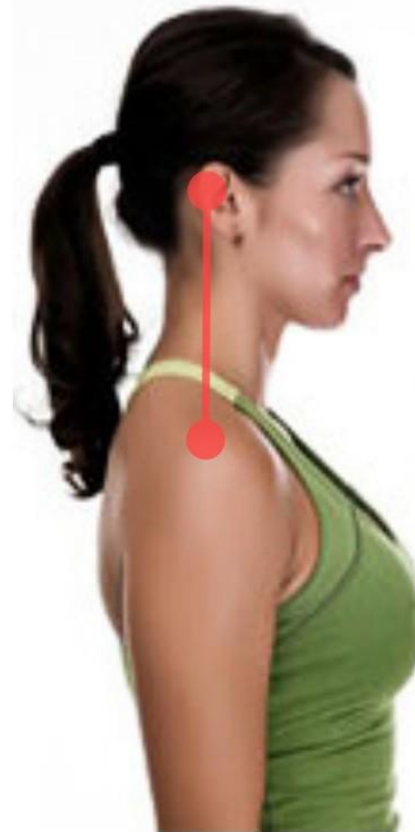
memorize

Posterior Postural Muscles	Posterior Phasic Muscles
Upper Cross	
Upper Trapezius	Mid-lower Trapezius
Levator Scapulae	Rhomboids
Cervical erector spinae	Latissimus dorsi
Lower Cross	
Quadratus lumborum	Gluteus minimus
Thoracolumbar erector Spinae	Gluteus medius
Hamstrings	Gluteus maximus
Piriformis	
Gastroc-soleus complex	



Clinical Association: HEAD FORWARD POSTURE and Upper Crossed Syndrome

- Increased thoracic kyphosis
Rounded shoulders
- Flexion of the lower cervical spine
- Extension of upper cervical spine
- Muscle imbalance –
tight/facilitated type I
muscles vs weak/inhibited
type II muscles



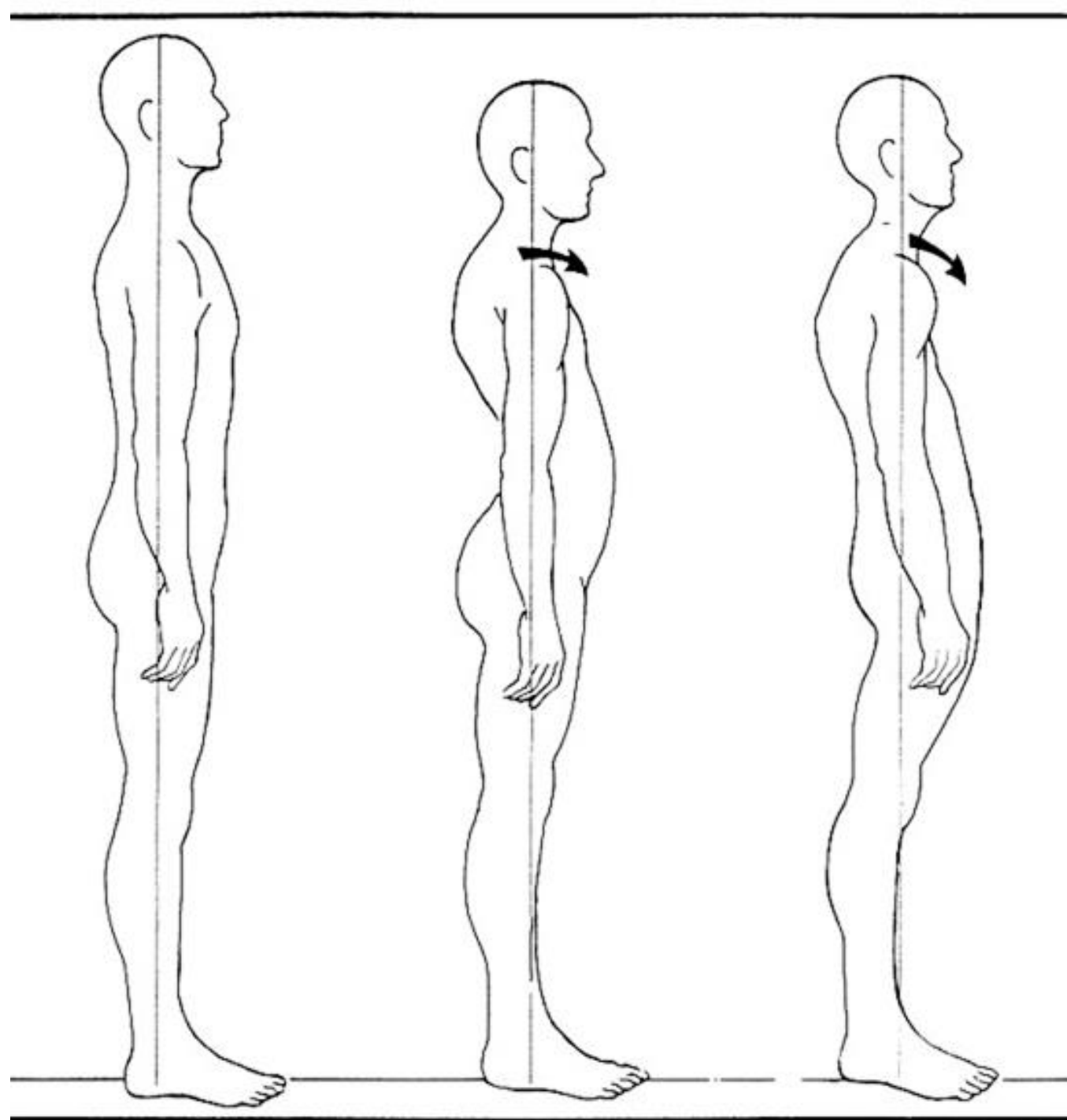
Correct Head Posture



Forward Head Posture

- Symptoms-compensatory or facilitative mechanisms may cause:

- Cervical, thoracic, lumbosacral pain
- GI disorders, abdominal, pelvic and inguinal pain
- Headache, neck pain



1

Scalenes



Stabilize clavicle using your opposite side hand. Tuck chin, translate head posteriorly. Side bend the neck away from affected side. Rotate either way into scalenes. Hold 15-20 seconds.

2

Levator Scapulae



grab edge of bed or chair. stabilize scapulae of the side with hypertonic muscle. Induce flexion, side bending, and rotation away from the scapulae, stretching into the barrier.. Feel a stretch, but not pain. This can also be done while lying on your back. Hold 15-20 seconds. Repeat on other side.

3

Sternocleidomastoid Upper Trapezius



synergistic with upper trapezius. Stabilize clavicle and first rib using your opposite hand. Induce flexion, side bending away and rotation toward the affected side. Stretch into the barrier. This can also be done while lying on your back. Hold 15-20 seconds. Repeat on other side.

Summary Slide

- The Upper Crossed Syndrome provides a clear framework from which to view a patient's postural compensations and subsequently target treatment approaches (*which can include an Exercise Prescription, as noted in the examples*).
- Other postural considerations are manifold and important to consider in a patient's clinical presentation.
- Application of the osteopathic perspective and the osteopathic philosophy can be beneficial for a patient's overall well being.